

Documents

Machine Learning (2023) 112:3977–4012 3995

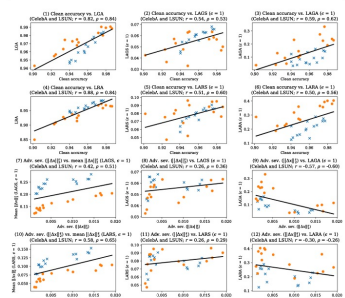


Fig. 3 Correlation plots for some of the data presented in Table 4 (plots 1–6, 8–9, 11–12) and additional data (plots 7 and 10). Plots are made for CelebA (blue crosses) and LSUN (orange circles) data combined (MNIST data is not shown). Plots 1 and 4 show the relation between clean accuracy LGA and LGA. Plots 2–3 and 5–6 show an association between clean accuracy and latent adversarial robustness (measured as LGS, LAGA, LARS, LARA). Plots 7 and 10 show an association between conventional adversarial robustness (measured as adversarial severity) with respect to perturbations bounded by scaled ℓ_2 norm and the averaged scaled ℓ_2 norm of found approximately minimum latent perturbations. Plots 8–9 and 11–12 demonstrate the inconclusive results regarding the association between conventional adversarial robustness and latent adversarial robustness. For each plot, Pearson's and Spearman's correlation coefficients (r and ρ , respectively) are given

the generative models were not trained to work. This can be explained by the simplicity of the MNIST classification problem.

Finally, we confirmed the meaning of decay in the latent space as a countermeasure against the increase of the norm of the latent vector by the adversary: as visible from Fig. 10, column 2, perturbed vectors typically exceed unperturbed vectors by norm. This phenomenon is explained by (1) the lower probability density of vectors with large latent space norms and the associated lack of classifier training on such less plausible input images; and (2) a higher ease to exploit a weakness of a generative model with the same sort of vectors. In particular, the second explanation applies to CelebA, where roughly half of approximately minimum latent space perturbations found with $\epsilon = 0.5$ ($d = 0.106$)

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Text-Only OCR

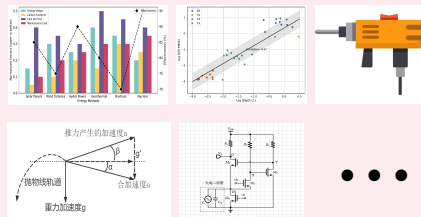
Multimodal OCR

Graphics

Text

Graphics

Graphics as Pixels



Often Discarded



Cropped images

Nature language



Text



Formula



Table

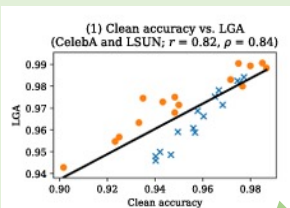
...

Markdown



Text tokens

Graphics as Code



Chart, Logos, Flow Diagrams Scientific Schematics ...

Parsed

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SVG Code

reconstruct

Ours



Image-Code Pair

Scale

Limited usage



Image Caption



VQA

Wider usage



Text2Image



Fine-grained Perception



Controllable Image Generation



Unified Generation & Understanding

Unstructured Data

Contrast in OCR Paradigms

Downstream Usage